

DE-LU Day-Ahead Power Price Forecasting — Case Study

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Market: Germany-Luxembourg (DE-LU) bidding zone, EPEX day-ahead, hourly, EUR/MWh | Forecast option: A — next-day hourly
| OOS window: 2025-01-01 → 2025-12-31 (full Test year, 8,759 predictions)

"I prioritised transparent, stable models deliberately: a fair-value signal has to be one a trader can interrogate and trust, and these markets break regime often enough that an over-flexible model is a liability, not an edge. The complex model is included as a benchmark — to confirm the simple one isn't leaving material signal behind."

This is a working prototype of a daily DE-LU fair-value view: next-day hourly forecasts, a prompt-curve relative-value lean, a trader-facing dashboard, and an LLM trader briefing, all from a single python `main.py` command. Full build history, round-by-round detail, and all figures are in `full_report.pdf`; this document covers the brief's required topics at summary depth.

1. Data, Sources & Point-in-Time Discipline

Target: EPEX DE-LU day-ahead hourly prices, 2019–2025 (history starts 2019 — the DE-AT-LU zone split in Oct 2018 mixed two price regimes before that). Source: ENTSO-E Transparency Platform — day-ahead (A01) **forecast** load/wind/solar for DE and FR, Forecast Transfer Capacity (NTC, the six borders that actually publish one), and neighbor bidding-zone prices/load/wind/solar (FR/NL/BE/PL/CZ). Plus daily TTF front-month gas (yfinance) and a CO2 proxy (CARB.L, a tradable ETC — no free official EUA settlement series was reachable, documented honestly rather than backfilled). EXAA's (Sequence 2) own day-ahead auction print for the same delivery day — already present unused in the committed price export — is used as a pre-auction reference (\$5), not a model feature.

Point-in-time firewall: every feature for delivery day D must be knowable by ~12:00 D-1 (gate closure). Load/wind/solar are explicitly the day-ahead *forecast* vintage, never realised — enforced in code by `assert_no_lookahead()`, which raises if any forbidden timestamp appears. Negative prices (3.3% of hours) are never clipped. DST 23h/25h days are handled via a tz-aware index.

2. Feature Engineering

Spine: **residual load = load – wind – solar** — the merit-order position, the dominant driver of both level and shape. 47 features total: calendar (cyclic hour/dow/month, holidays), residual load + hinge/quadratic merit-order terms, D-1/D-7 price lags (DE and 5 neighbor zones — in LightGBM's gain ranking, the CZ and BE lags actually outrank DE's own), rolling residual-load and price stats, TTF gas anchor (+ a Ridge-only gas×residual-load interaction), a standardised gas+CO2 pressure composite, FR residual load, NTC import/export/net capacity, and neighbor residual loads. Generation/grid outage data was considered and explicitly dropped — pre-2021 ENTSO-E REMIT data is sparse and unstructured, and forcing it in risked fabricating precision the rest of this build avoids.

3. Model Lineup & Selection

Baselines: D-1 and D-7 same-hour naïve. **Selected: Ridge** with hinge/quadratic residual-load terms — plain linear mis-specifies the convex, kinked merit-order curve. **Challenger: LightGBM**, included only as a nonlinearity check. A depth-3 decision tree is kept as an interpretability figure (not selected — high-bias, blocky), since its splits literally draw the merit-order kink.

Model	MAE — Validation 2024 (expanding)
Ridge (selected)	14.32
LightGBM (challenger)	11.22

LightGBM posts a lower Validation MAE (gap 3.10 / 21.7%), yet **Ridge is selected**: (1) a fair-value signal must be interrogable — coefficients vs. a 500-tree ensemble; (2) power markets break regime, and a regularised linear form degrades more gracefully than a flexible model when the next regime looks nothing like 2019-2024; (3) LightGBM's feature importances confirm Ridge's design (price lags, residual load dominate) rather than showing it's mis-specified. This decision is made on Validation (2024) only — the Test (2025) gap (LightGBM ahead by 3.27, 22.6%) is reported only as out-of-sample confirmation, never as part of the choice.

4. Validation Methodology & Test Results

Three-way split: Train 2019-2023 (fit), Validation 2024 (window-type + model-selection decision, used once), Test 2025 (headline backtest, never tuned on). Walk-forward throughout: train on all valid rows strictly before day D, predict D's 24 hours, advance — no shuffling. Calibration window (expanding vs. rolling) is chosen on Validation MAE alone; expanding currently wins (14.32 vs. 15.05). No MAPE is reported — DE-LU prices cross zero, so it's undefined/explosive; MAE and RMSE are used throughout.

Model	MAE	RMSE	Skill vs. D-7
Naïve D-7	32.83	49.33	+0.0%
Naïve D-1	25.97	40.45	+20.9%
Ridge (selected)	14.50	23.58	+55.8%
LightGBM (challenger)	11.23	18.93	+65.8%

Test 2025, 8,759 hourly predictions (EUR/MWh). By regime, Ridge/LightGBM MAE: peak 18.18/14.82, off-peak 12.17/8.97, high residual load 14.77/11.67, low residual load 14.23/10.79. `predictions.csv` covers the full Test year (deliberately widened from the brief's "–2-4 weeks" so the deliverable matches the headline backtest it represents — see §6).

5. Prompt-Curve Translation

A next-day model cannot produce a month-ahead curve — rolling a 1-day forecast 30 days forward compounds error against driving forecasts that don't exist. The view instead has two honest parts: (a) the model's **own hourly/block shape** — peak 88.05, off-peak 80.24 EUR/MWh (spread +7.80), 2,902 rich / 2,472 cheap hours (incl. 442 negative) / 876 scarcity hours — directly tradable with no external reference; and (b) a **directional call vs. EXAA**, the Sequence-2 pre-auction print for the same delivery day (full-history correlation with the EPEX target: 0.986). Conviction/size scale with `[basis] ÷ model backtest MAE`, not an arbitrary EUR threshold. Worked example (2025-12-31): EXAA 84.00, Ridge fair value 76.49 → basis –7.51 (0.52× MAE) → SELL, MODERATE conviction.

6. LLM Trader Commentary

A grounded LLM (Anthropic API) turns the structured model state into a trader-facing morning note. It **originates no numbers** — every quantity traces to a structured fact object; output is schema-validated (Pydantic v2: bullish/bearish drivers, invalidation triggers, narrative text), and a grounding check verifies every number in the prose against the fact object. Direction/conviction are *not* LLM outputs — they're pure arithmetic (§7) the LLM only narrates. Cached to `ai_logs/` so `python main.py` reproduces deterministically, key-free.

7. Fair-Value-vs-EXAA Dashboard

`outputs/fair_value_dashboard.html` is a trader-facing, fully offline page (Chart.js vendored locally, no CDN) replaying one delivery day as a **static backtest** from the committed snapshot. The decision layer is pure arithmetic, computed once in `src/dashboard.py` and shared with the LLM's fact object so the two can never disagree: per hour, $\text{basis}_h = \text{fair_value}_h - \text{EXAA}_h$, $\text{conviction}_h = |\text{basis}_h| \div \text{MAE}_h$ (that hour's own backtest MAE), direction FLAT below a 0.25 conviction threshold else SELL/BUY by sign — rolled up into baseload/peak/off-peak block calls. For 2025-12-31: 22 SELL / 2 FLAT / 0 BUY across the 24 hours. Every number on the page is verified against its source JSON by an automated grounding check that fails the build loudly if anything doesn't trace back — the realised EPEX settlement was in fact removed after this check (and a correctness review) flagged it as a look-ahead violation, since it isn't known until after the auction the page exists to help decide.

*Full rationale, round-by-round build history, all figures, and the complete LLM/dashboard design discussion are in `full_report.pdf`.
Reproduction: `python main.py` (single command, ~27s on a cache-hit run, no API keys required, deterministic).*